BeforeYouSign

MINOR PROJECT REPORT

Submitted in partial fulfillment of the requirement for the Degree of Bachelors of Engineering in Computer Science & Engineering

Submitted To:

[PARUL UNIVERSITY, VADODARA, GUJARAT (INDIA)]

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DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

PARUL INSTITUTE OF TECHNOLOGY VADODARA, GUJARAT

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DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

CERTIFICATE

This is to certify that **Srinivas Anjaneyulu Jangiti, Vineet Krishna Gorule, Amit Bansilal Sahani,** Students of **CSE VI Semester** of **" Parul Institute of Technology, Vadodara"** has completed their **Minor Project** titled **"BEFOREYOUSIGN"**, as per the syllabus and has submitted a satisfactory report on this project as a partial fulfillment towards the award of degree of **Bachelor of Technology** in **Computer Science and Engineering** under **Parul University, Vadodara, Gujarat (India)**.

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DECLARATION

We the undersigned solemnly declare that the project report "**BEFOREYOUSIGN"** is based on my own work carried out during the course of our study under the supervision of **PROF. VINOD KUMAR YADAV, Assistant Professor, CSE**.

We assert the statements made and conclusions drawn are the outcomes of my own work. I further certify that

1. The work contained in the report is original and has been done by us under the general supervision of our supervisor.
2. The work has not been submitted to any other Institution for any other degree / diploma / certificate in this university or any other University of India or abroad.
3. We have followed the guidelines provided by the university in writing the report.

Whenever we have used materials (data, theoretical analysis, and text) from other sources, we have given due credit to them in the text of the report and giving their details in the references.

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ACKNOWLEDGEMENT

In this semester, we have completed our project on **"BEFOREYOUSIGN"**. During this time, all the group members collaboratively worked on the project and learnt about the industry standards that how projects are being developed in IT Companies. We also understood the importance of teamwork while creating a project and got to learn the new technologies on which we are going to work in the near future.

We gratefully acknowledge for the assistance, cooperation guidance and clarification provided by **"PROF. VINOD KUMAR YADAV"** during the development of our project. We would also like to thank our Head of Department **Prof. Sumitra Menaria** and our Principal **Dr. Swapnil Parikh** Sir for giving us an opportunity to develop this project. Their continuous motivation and guidance helped us overcome the different obstacles for completing the Project.

We perceive this as an opportunity and a big milestone in our career development. We will strive to use gained skills and knowledge in our best possible way and we will work to improve them.

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LIST OF ABBREVIATIONS

S. No.	Abbreviation	Full Form
1	AI	Artificial Intelligence
2	ML	Machine Learning
3	NLP	Natural Language Processing
4	LLM	Large Language Model
5	BERT	Bidirectional Encoder Representations from Transformers
6	API	Application Programming Interface
7	JWT	JSON Web Token
8	OCR	Optical Character Recognition
9	EVM	Ethereum Virtual Machine
10	CUAD	Contract Understanding Atticus Dataset
11	RLS	Row-Level Security
12	SSE	Server-Sent Events
13	DFD	Data Flow Diagram
14	ER	Entity Relationship
15	GUI	Graphical User Interface
16	UML	Unified Modeling Language
17	RAM	Random Access Memory
18	GB	Gigabyte
19	IDE	Integrated Development Environment
20	NDA	Non-Disclosure Agreement
21	SME	Small and Medium Enterprise

ABSTRACT

BeforeYouSign is an online AI-powered legal contract intelligence platform developed to eliminate the information asymmetry that disadvantages individuals, freelancers, and small businesses when signing commercial contracts. The main objective of this system is to provide a fast, accessible, and easy-to-understand contract risk analysis service for users who cannot afford professional legal review. The platform allows any user to upload a contract document in PDF, DOCX, or image format and receive a comprehensive clause-level risk report within 30 seconds, making institutional-grade legal analysis available to everyone.

The platform integrates several intelligent features. The AI Contract Analyzer identifies high-risk clauses and explains them in plain language. The Negotiation Engine generates specific counter-proposals for each risky clause, calibrated to the user's leverage position. The ML Risk Predictor uses a machine learning model trained on over 12,000 annotated contract clauses to score the dispute probability of each clause. The Market Benchmark Engine compares contract terms against a database of 5,000+ real commercial contracts. The Obligation Tracker automatically extracts all contractual deadlines and sends timely reminders. The Blockchain Registry provides tamper-evident cryptographic registration of executed contracts on Ethereum or Polygon. The Lawyer Marketplace connects users with verified legal professionals for complex matters.

BeforeYouSign is built using the Next.js full-stack framework with TypeScript, Supabase PostgreSQL as the database with Row-Level Security for data privacy, and is deployed on the Vercel Edge Network for global availability. As of March 2026, the platform has analyzed over 10,000 contracts for more than 5,000 users across 37 countries, maintaining analysis latency below 30 seconds at the 95th percentile and a platform uptime of 99.9%.

Overall, BeforeYouSign improves the legal literacy of non-institutional contract parties by combining AI-powered analysis, machine learning risk prediction, and intelligent contract management tools in a single, user-friendly web application.

Keywords: Legal Technology, Artificial Intelligence, Machine Learning, Natural Language Processing, BERT, Contract Analysis, Risk Detection, Negotiation Engine, Blockchain Verification, BeforeYouSign

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CHAPTER I INTRODUCTION

Project Title :

BeforeYouSign – AI-Powered Legal Contract Intelligence Platform for Risk Analysis and Negotiation Support.

1.1 Overview :

BeforeYouSign is an innovative online platform developed to provide AI-powered contract analysis and legal intelligence services to individuals, freelancers, startups, and small businesses. The platform aims to eliminate the information asymmetry that arises when non-institutional parties sign commercial contracts without understanding the legal implications of the clauses they are agreeing to. Professional legal contract review typically costs between USD 700 and USD 2,400 per contract in major markets, making it inaccessible for the vast majority of individuals and small businesses who routinely sign employment agreements, non-disclosure agreements, vendor contracts, software subscriptions, and lease agreements.

In addition to providing instant contract risk analysis, BeforeYouSign enhances the contract management experience by integrating several intelligent features including AI-based clause explanation in plain language, machine learning-based dispute probability scoring, negotiation strategy generation with counter-proposals, market benchmarking against real commercial contracts, automated obligation tracking with deadline reminders, and cryptographic contract verification on blockchain networks. These features work together to help users identify risks, negotiate better terms, and manage their contractual obligations more effectively.

The platform ultimately aims to make legal contract comprehension more intelligent, convenient, and accessible for all users, while also giving small businesses the tools to negotiate more equitably with larger institutional counterparties.

1.2 Problem Statement :

The traditional contract review process faces several significant challenges that reduce its accessibility, efficiency, and fairness for non-institutional parties. Many individuals, freelancers, and small businesses sign legal contracts without fully understanding the implications of every clause, primarily because professional legal review is too expensive and time-consuming for routine contractual decisions.

A major issue is that legal contracts are written in complex, technical language that is difficult for non-lawyers to understand. Clauses related to intellectual property assignment, non-compete obligations, automatic renewal terms, unlimited indemnification, and governing law are commonly misunderstood or overlooked, leading to unfavourable outcomes after signing. The World Justice Project (2023) reports that in 113 of 142

surveyed countries, the majority of the population cannot access affordable legal assistance, highlighting the global scale of this problem.

Furthermore, there is currently no centralized digital platform that combines contract risk identification with negotiation strategy, obligation management, and blockchain verification in a single accessible tool. Users usually need to consult multiple different sources to understand a contract, find a lawyer, track deadlines, and verify document integrity. This fragmented system makes contract management inconvenient, expensive, and inaccessible for many people.

Therefore, there is a strong need for a modern AI-powered digital solution that analyzes contracts instantly, explains risks in plain language, provides negotiation guidance, tracks obligations automatically, and verifies documents cryptographically — all in one platform accessible to anyone with a web browser.

1.3 Objective of Project :

The main objective of the BeforeYouSign platform is to develop an AI-powered legal contract intelligence system that makes professional-grade contract analysis accessible to everyone. The platform aims to:

- Provide a user-friendly online platform where users can upload any contract document in PDF, DOCX, or image format and receive a comprehensive risk analysis report within 30 seconds.
- Decode complex legal language and clause structures into clear, plain-language explanations that any non-lawyer can understand without sacrificing legal accuracy.
- Implement ML-based dispute probability scoring using a fine-tuned BERT transformer model trained on over 12,000 annotated contract clauses, providing an independent statistical risk signal alongside the AI analysis.
- Provide AI-generated negotiation recommendations and specific counter-proposal language for each high-risk clause, calibrated to the user's negotiating leverage position.
- Compare contract terms against a database of 5,000+ real commercial contracts through semantic vector search, showing users whether their terms are above or below market average.
- Automatically extract all contractual obligations including deadlines, payment schedules, notice periods, and renewal windows, and send automated email and SMS reminders at configurable lead times.
- Provide cryptographically tamper-evident registration of executed contracts on EVM-compatible blockchain networks (Ethereum and Polygon), generating a verification certificate for evidentiary purposes.
- Connect users with verified jurisdiction-specific legal professionals for complex matters that require qualified human legal counsel.

1.4 Application or Scope :

The scope of the BeforeYouSign platform is to develop a comprehensive AI-powered legal contract intelligence system that serves individuals, freelancers, startups, and small businesses across all industries and jurisdictions. The platform addresses the full lifecycle of a commercial contract from pre-signature analysis through execution, obligation management, and cryptographic verification.

Through this platform, users will be able to upload and receive detailed risk analysis for employment agreements, non-disclosure agreements, vendor and supplier contracts, software-as-a-service subscriptions, lease deeds, partnership agreements, and general commercial contracts. The system provides per-clause risk scores, plain-language explanations, negotiation strategies, and market comparisons to help users make informed decisions before signing.

Users will be able to:

- Upload contracts in PDF, DOCX, PNG, or JPG format and receive analysis in under 30 seconds.
- View clause-level risk scores with severity ratings (Critical, High, Medium, Low) and plain-language explanations.
- Get AI-generated negotiation counter-proposals calibrated to their leverage position.
- Compare their contract terms against industry norms using the market benchmark engine.
- Track all contractual obligations with automated email and SMS deadline reminders.
- Register executed contracts on blockchain for tamper-evident cryptographic verification.

The platform supports contract analysis in over 50 languages and is optimized for common-law jurisdictions including India, the United States, the United Kingdom, and Australia. An admin control system monitors platform activity, manages users, and maintains system security. Overall, BeforeYouSign aims to create a centralized digital ecosystem for contract intelligence, improving legal accessibility for all users.

1.5 Organization of the Report :

This report is organized into several chapters to explain the development of the BeforeYouSign system comprehensively.

Chapter 1 presents the introduction, including the project overview, problem statement, objectives, scope, and organization of the report.

Chapter 2 discusses the literature survey, covering a review of existing AI-based legal analysis systems, prior research on contract understanding datasets, and a comparison of existing platforms with BeforeYouSign.

Chapter 3 explains the methodology used in the project, including the system architecture, project platforms, proposed end-to-end analysis pipeline, module descriptions, and all system diagrams including ER Diagram, Use Case Diagram, and Data Flow Diagrams.

Chapter 4 describes the system requirements including both software and hardware requirements for the development and production environments.

Chapter 5 presents the expected outcomes of the project including quantitative performance targets, qualitative societal impact, and GUI screenshots illustrating the platform's user interface.

Chapter 6 provides the conclusion of the project and a structured future work roadmap covering planned improvements and new features.

Chapter 7 lists all references in the prescribed format including research papers, books, and web resources used in this project.

CHAPTER II LITERATURE SURVEY

The literature survey focuses on studying existing systems, technologies, and research related to AI-powered legal text analysis, contract understanding, and intelligent legal technology platforms. With the advancement of machine learning and natural language processing technologies, several researchers and organizations have developed systems that attempt to automate the analysis of legal documents. However, most existing systems either focus narrowly on one specific task such as clause classification, or are restricted to specific contract types and jurisdictions, or lack the integrated workflow that makes legal analysis practically usable for non-institutional users.

2.1 Existing AI-Based Legal Analysis Systems

2.1.1 CLAUDETTE (Lippi et al., 2019)

CLAUDETTE is an automated detection system for potentially unfair clauses in online Terms of Service documents, developed by researchers at the University of Bologna. The system uses Support Vector Machines (SVMs) and Convolutional Neural Networks (CNNs) to classify clauses from online Terms of Service across eight unfairness categories including unilateral termination, price change, and limitation of liability. On a corpus of 50 annotated Terms of Service documents, CLAUDETTE achieved precision above 80% and recall above 73%. This work established the technical feasibility of automated unfairness detection in legal text and directly inspired the multi-class risk classification approach used in BeforeYouSign.

Limitation: CLAUDETTE is limited to online Terms of Service and does not support commercial contracts, negotiation guidance, or any integrated workflow for users.

2.1.2 CUAD — Contract Understanding Atticus Dataset (Hendrycks et al., 2021)

Hendrycks et al. (2021) published the Contract Understanding Atticus Dataset (CUAD), an expert-annotated NLP benchmark containing 510 commercial contracts annotated across 41 legally significant clause categories by qualified legal professionals. Models fine-tuned on CUAD, particularly RoBERTa-large, achieved a mean precision of 87% across all 41 categories on the held-out test set. CUAD established a reliable taxonomy for commercial contract clause classification and became the standard benchmark for evaluating contract analysis systems. BeforeYouSign's ML Risk Predictor is fine-tuned on the CUAD training set augmented with 8,500 proprietary annotations, and its risk taxonomy is directly based on CUAD's 41 clause categories with 8 additional proprietary categories.

Limitation: CUAD models perform clause extraction and classification but do not provide plain-language explanations, negotiation recommendations, or any integrated user-facing features.

2.1.3 GPT-4 and Large Language Model Legal Reasoning (Bommarito and Katz, 2022)

Bommarito and Katz (2022) demonstrated that GPT-4 achieved a Uniform Bar Examination pass rate of 75.7%, equivalent to the 90th percentile of human examinees,

without any task-specific fine-tuning. This result established that frontier LLMs possess the breadth of legal knowledge and reasoning capability to analyze novel legal questions including contract clause interpretation at a level previously achievable only by qualified attorneys. BeforeYouSign uses GPT-4o as its primary AI inference engine, which extends GPT-4's legal reasoning capability with a 128,000-token context window, native JSON structured output mode, and multimodal input for scanned document images.

Limitation: LLM-only legal analysis lacks the statistical grounding of supervised ML classification. BeforeYouSign addresses this by combining both approaches — using an ML classifier for statistically grounded risk scores and an LLM for qualitative explanation and negotiation guidance.

2.2 Existing Legal Technology Platforms

Several commercial legal technology platforms currently offer contract analysis services:

Kira Systems and Luminance are AI-powered contract analysis tools targeted at large law firms and enterprise customers. They perform clause extraction and data extraction for due diligence, but are priced exclusively for institutional users and do not provide plain-language explanations, negotiation strategies, obligation tracking, or blockchain verification for individual users.

LawGeex and Ironclad offer contract review automation but are designed as enterprise CLM (Contract Lifecycle Management) tools requiring significant setup and integration effort, making them unsuitable for individuals or small businesses needing on-demand analysis.

Limitations of existing systems: Most existing platforms focus on enterprise customers at enterprise prices, do not combine ML scoring with LLM explanation, do not provide negotiation counter-proposals, do not track obligations, and do not offer blockchain verification. BeforeYouSign is designed specifically to serve the underserved individual and SME market with a free-to-start, web-based platform accessible to anyone.

2.3 Comparative Analysis of Prior Systems

Capability	CLAUDETTE (2019)	CUAD (2021)	GPT-4 (2022)	Commercial LegalTech	BeforeYouSign (2026)
Unfair/Risk Clause Detection	Yes (80%+)	Yes (87%+)	Yes (0-shot)	Yes (varies)	Yes (82–95%)
Plain-Language Explanation	No	No	Partial	Partial	Yes (structured)
Negotiation Counter-Proposals	No	No	No	No	Yes (clause-level)
Market Benchmarking	No	No	No	Partial	Yes (5,000+ contracts)
Obligation Tracking & Reminders	No	No	No	Partial	Yes (auto-extracted)

Blockchain Verification	No	No	No	No	Yes (ETH + Polygon)
Free / Accessible Pricing	N/A	N/A	N/A	No (enterprise)	Yes (free tier)
Production Deployment Scale	No	No	No	Yes (enterprise)	Yes (10,000+ contracts)

[Table 1 : Literature Survey Comparison Matrix]

CHAPTER III METHODOLOGY

3.1 Background / Overview of Methodology :

The development of the BeforeYouSign platform follows a structured methodology to design, implement, and deploy a reliable AI-powered contract intelligence system. The methodology focuses on understanding user requirements, designing the system architecture, developing the analytical pipeline, implementing all functional modules, and testing system performance comprehensively.

Initially, requirement analysis was performed to identify the challenges faced by individuals and small businesses when dealing with legal contracts. After this, the system design phase was carried out, where the database structure, system modules, and UML diagrams including the Data Flow Diagram, Entity-Relationship Diagram, and Use Case Diagram were prepared. The platform is developed using the Next.js full-stack framework with TypeScript, Supabase PostgreSQL with the pgvector extension as the database, and OpenAI GPT-4o and Anthropic Claude 3.5 Sonnet as the primary AI inference engines. Finally, the system is tested and deployed on the Vercel Edge Network to ensure smooth operation and efficient performance.

3.2 Project Platforms Used in Project :

Frontend Technologies

- Next.js 14 (App Router) — Used for building the full-stack web application with server-side rendering, React Server Components for reduced client bundle size, and Server-Sent Events (SSE) for progressive streaming of analysis results to the user interface.
- React 18 — Used for building reusable, dynamic user interface components with efficient state management.
- TypeScript 5 — Used to add strict static type checking across all frontend and backend code, catching bugs at compile time and ensuring reliable API contracts.
- Tailwind CSS + shadcn/ui — Used for styling and designing the user interface with accessible, WCAG 2.1 compliant UI components.

Backend Technologies

- Next.js API Routes (Edge Runtime) — Used for building serverless API endpoints with JWT authentication middleware and per-route rate limiting via Upstash Redis.
- LangChain.js — Used for orchestrating multi-step AI workflows, managing prompt templates, and parsing structured outputs from LLM API responses.

AI and Machine Learning

- OpenAI GPT-4o — Used as the primary large language model for clause interpretation, plain-language explanation generation, and negotiation counter-proposal drafting. Provides a 128,000-token context window and native JSON structured output mode.

- Anthropic Claude 3.5 Sonnet — Used as the secondary LLM fallback with a 200,000-token context window, providing independent second opinions on critical-severity clauses.
- Fine-tuned BERT-base Transformer — Used as the ML Risk Classifier for statistically grounded dispute probability scoring per clause, deployed on a dedicated NVIDIA A10G GPU inference endpoint.

Database and Storage

- Supabase / PostgreSQL 15 — Used for relational data storage with Row-Level Security (RLS) enforced at the database engine layer, ensuring complete per-user data isolation. The pgvector extension enables semantic similarity search for market benchmarking.
- Supabase Storage (AWS S3-backed) — Used for secure storage of uploaded contract documents with AES-256-GCM encryption at rest.

Blockchain

- Ethers.js v6 + Alchemy RPC — Used for EVM-compatible interaction with Ethereum and Polygon mainnet networks, enabling SHA-256 document hash registration for tamper-evident contract verification.

Deployment

- Vercel Edge Network (Pro) — Used for deploying the web application on a global Content Delivery Network with 100+ Points of Presence, automatic horizontal scaling, and sub-50ms cold start latency.

3.3 Proposed Methodology :

The proposed BeforeYouSign system integrates an AI-powered contract analysis engine with intelligent contract management tools. It allows users to upload and analyze any legal contract while also receiving structured negotiation assistance, obligation tracking, and cryptographic verification in one unified platform.

The system workflow begins when a user uploads a contract document. The document passes through an eight-stage automated analysis pipeline: (1) Ingestion and SHA-256 hashing; (2) Format-specific text extraction using PDF.js, Tesseract OCR, or mammoth.js; (3) Structural parsing to identify sections and clause boundaries using regex and BERT classifiers; (4) Clause segmentation and CUAD taxonomy tagging; (5) Dense embedding generation for semantic search; (6) ML Risk Scoring via the fine-tuned BERT classifier running in parallel; (7) LLM Analysis via GPT-4o for high-risk clauses running in parallel; (8) Market benchmarking against the precedent corpus and report assembly, streamed progressively to the user via Server-Sent Events.

Sellers and administrators manage the platform through role-specific interfaces. Verified lawyers receive client referrals through the marketplace module. This integrated workflow improves the efficiency of contract review and provides a convenient, intelligent, and accessible legal intelligence platform for all users.

3.4 Project Modules :

The BeforeYouSign system is divided into seven functional modules to ensure efficient development, easy maintenance, and better system organization. Each module is responsible for handling a specific set of tasks within the system.

After analyzing the system requirements and user roles, the following major modules have been finalized for the project:

- AI Contract Analyzer Module
- AI Negotiation Engine Module
- ML Risk Predictor Module
- Market Benchmark Engine Module
- Obligation Tracker Module
- Blockchain Registry Module
- Lawyer Marketplace Module

Each module manages a specific set of activities in the system. The AI Contract Analyzer Module handles document upload and the complete analysis pipeline. The AI Negotiation Engine Module generates counter-proposals and leverage-calibrated negotiation strategy. The ML Risk Predictor Module provides statistically grounded dispute probability scores independent of the LLM. The Market Benchmark Engine Module compares contract terms against real commercial precedents. The Obligation Tracker Module extracts and manages all contractual deadlines with automated reminders. The Blockchain Registry Module

handles cryptographic document hash registration. The Lawyer Marketplace Module connects users with verified legal professionals. Dividing the system into modules makes the project easier to design, implement, test, and maintain.

Explanation of Module-wise Functionality :

The BeforeYouSign system is divided into multiple modules, each designed to perform specific operations within the platform. These modules work together to ensure smooth functioning of the contract analysis and management system.

Name of System: BeforeYouSign

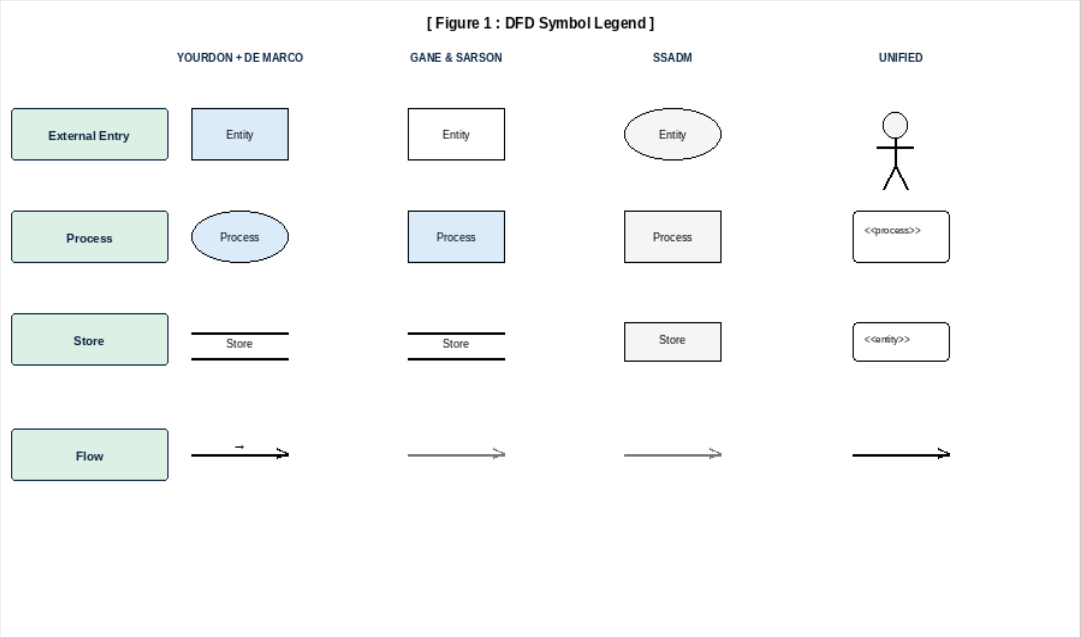
Assumptions: Users are already registered and logged in.

3.5 Diagrams (ER, Use Case, DFD, etc.) :

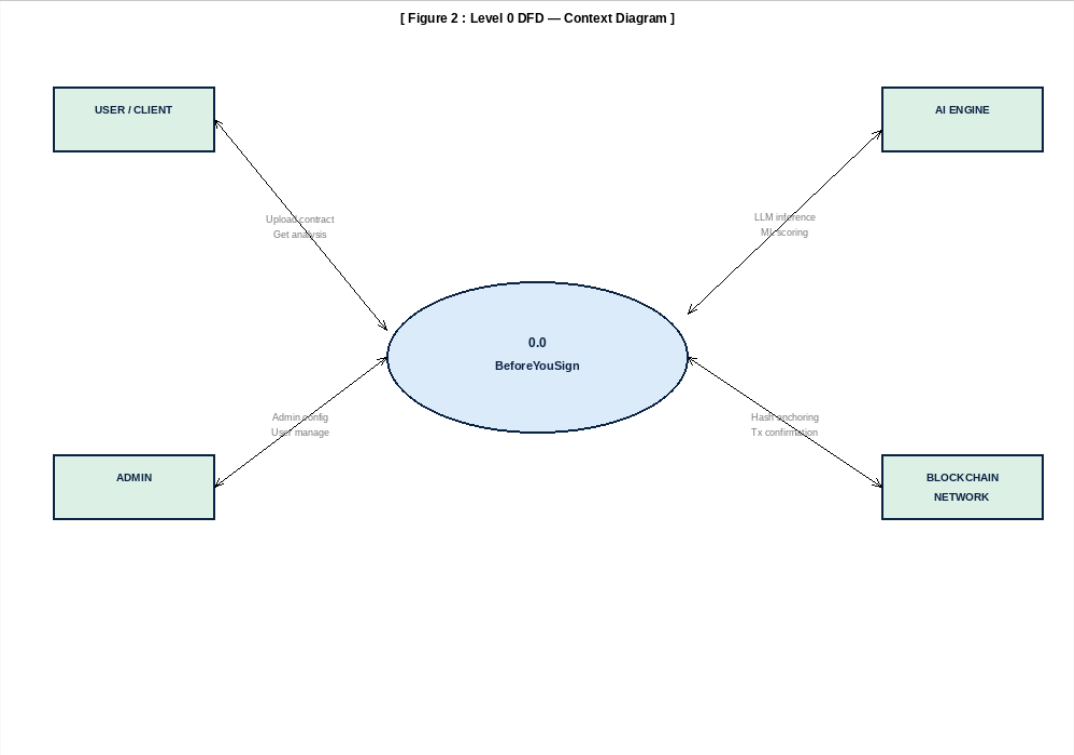
System design and process flow :

The BeforeYouSign system is designed to help users analyze, understand, and manage legal contracts efficiently. It includes a user interface, an eight-stage AI processing pipeline, and a database system to store contract data, analysis results, and user information. The system allows users to upload contract documents and receive a comprehensive risk analysis report within 30 seconds.

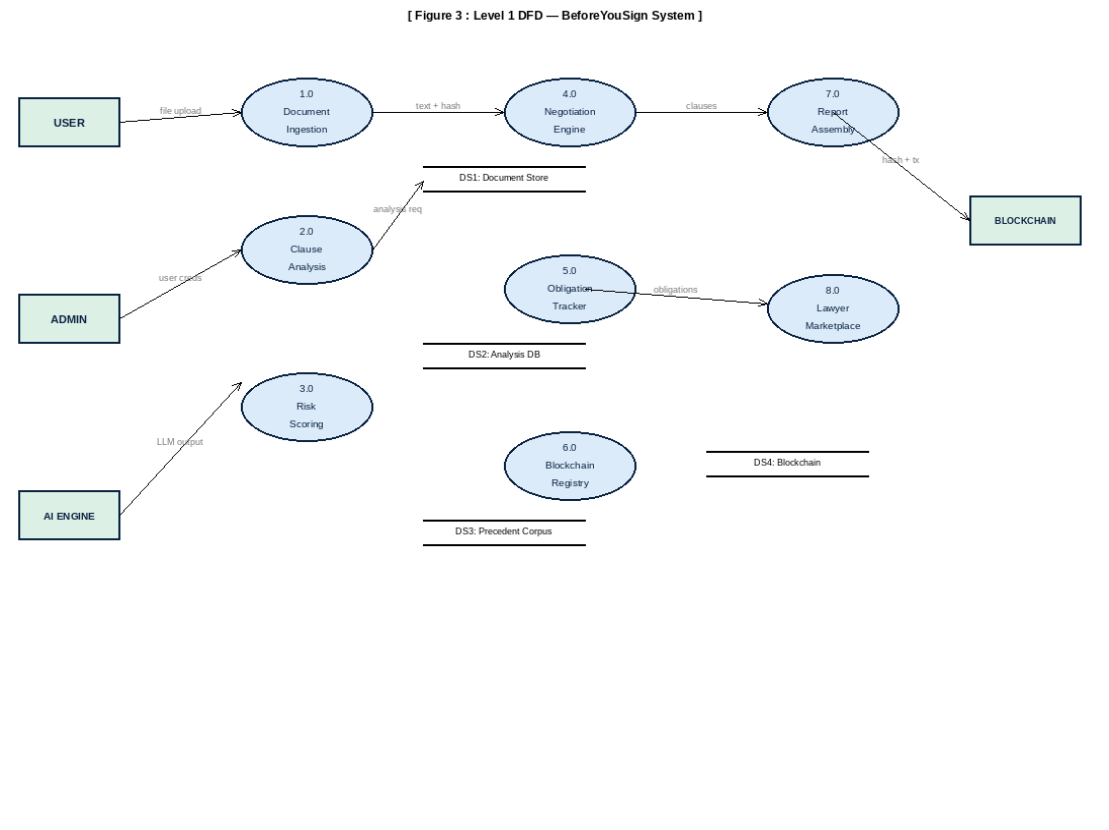
[Fig. 1 : DFD Symbol Legend]



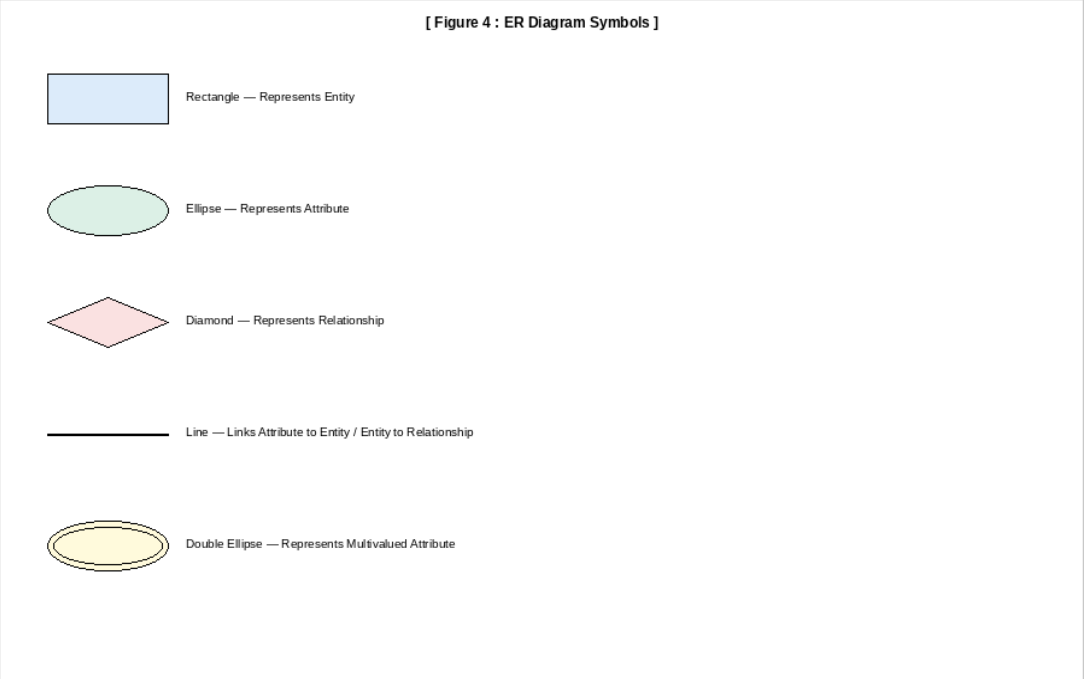
[Fig. 2 : Level 0 DFD (Context Diagram)]



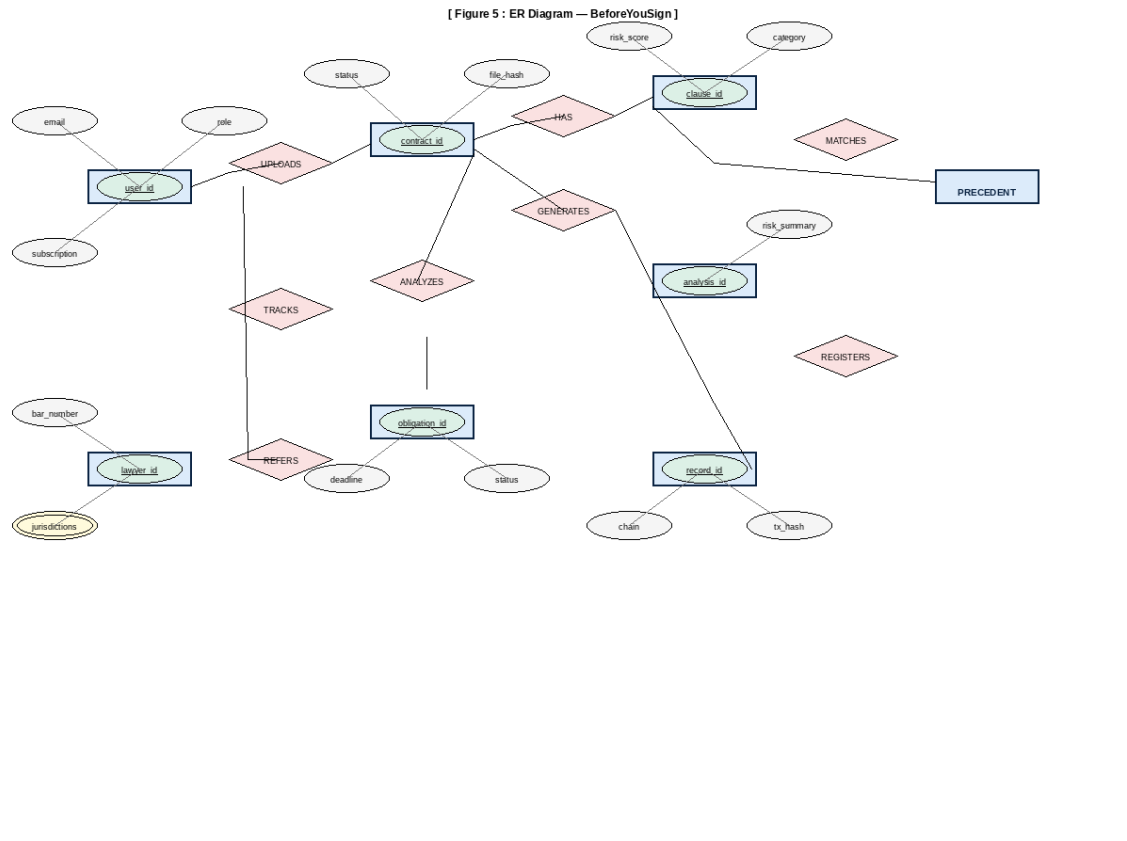
[Fig. 3 : Level 1 DFD — BeforeYouSign System]



[Fig. 4 : ER Diagram Symbol]

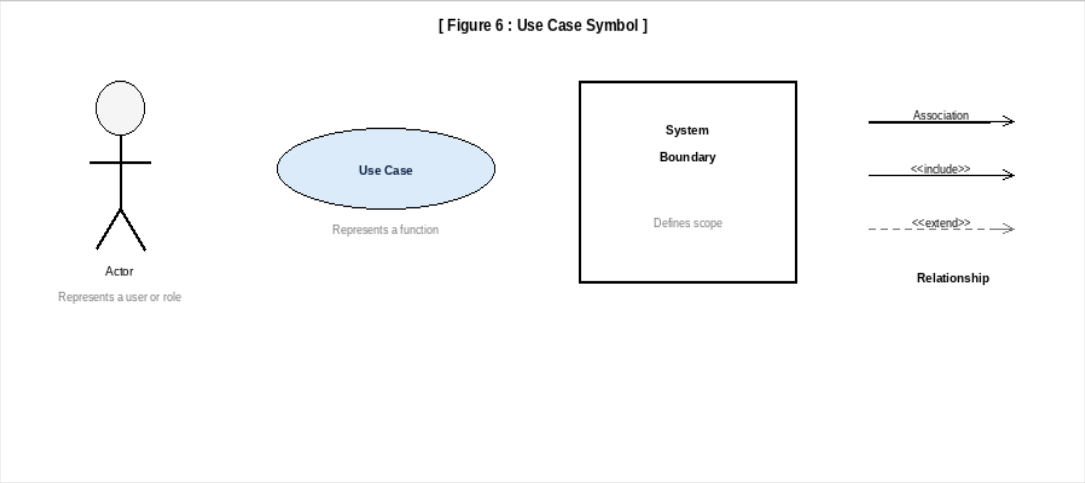


[Fig. 5 : ER Diagram — BeforeYouSign]

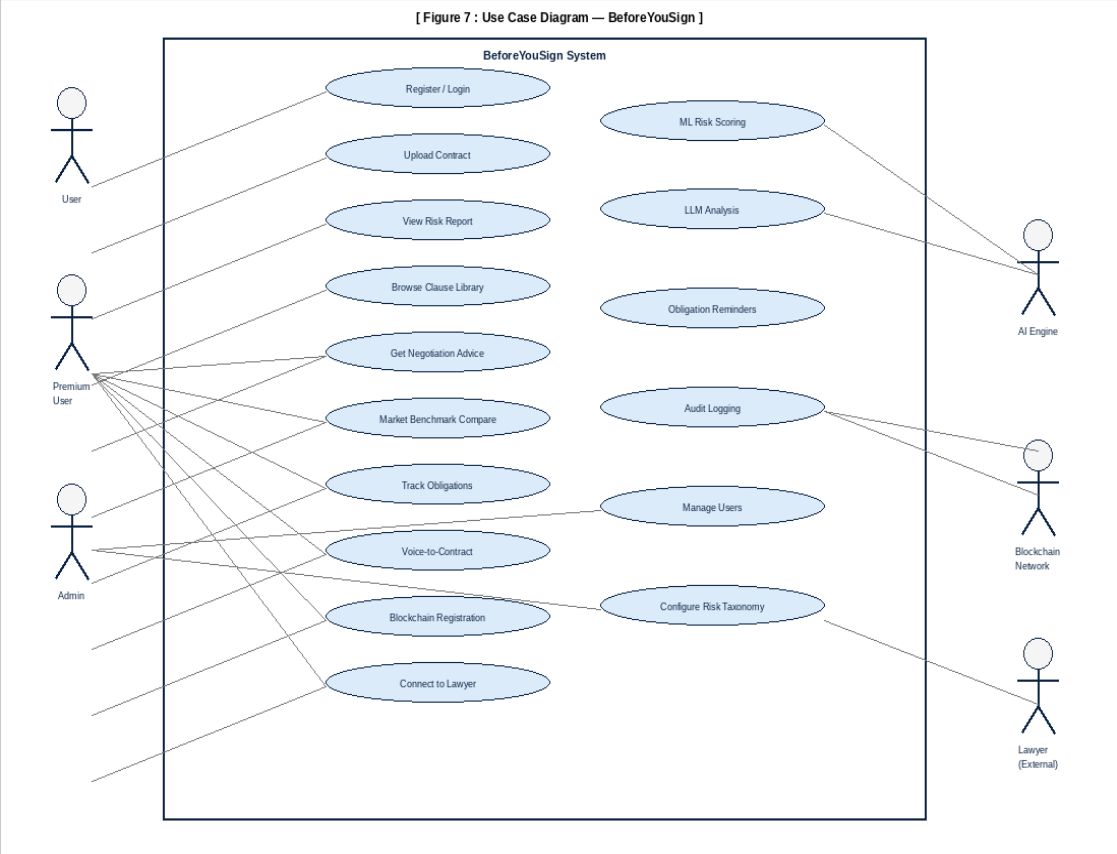


USE CASE DIAGRAM :

[Fig. 6 : Use Case Symbol]



[Fig. 7 : Use Case Diagram]



ML Risk Classifier Performance :

The ML Risk Predictor uses a fine-tuned BERT-base transformer model trained on 12,400 annotated contract clauses across 49 clause categories. The following table shows its performance on the held-out test set:

Contract Type	Train Samples	Precision	Recall	F1 Score	Accuracy
Employment Agreements	2,400	96.1%	94.2%	95.1%	95%
Non-Disclosure Agreements	3,100	94.3%	91.8%	93.0%	93%

Software / SaaS Agreements	1,800	90.2%	88.1%	89.1%	89%
Freelance / Service Contracts	1,200	88.7%	85.4%	87.0%	87%
Lease & Property Agreements	1,400	88.9%	84.9%	86.8%	87%
General Commercial Contracts	2,500	83.5%	80.7%	82.1%	82%
Weighted Average (all types)	12,400	90.3%	87.5%	88.9%	**87%**

[Table 3 : ML Risk Classifier Performance Metrics by Contract Type]

CHAPTER IV SYSTEM REQUIREMENTS

Project Title : BeforeYouSign – AI-Powered Legal Contract Intelligence Platform

4.1 Software Requirements :

The following software technologies and tools are required for the development and execution of the BeforeYouSign system:

S.No.	Software / Technology	Version	Purpose
1	**Operating System**	Windows 10+, Linux, macOS	Platform for running development tools and web browser
2	**Next.js (Full-Stack Framework)**	14.2.x	Server-side rendering, API routes, streaming SSE
3	**TypeScript**	5.4.x	Type-safe full-stack development; Zod for runtime schema validation
4	**Tailwind CSS + shadcn/ui**	3.4.x	Responsive UI design; WCAG 2.1 accessible components
5	**Supabase / PostgreSQL**	15 (Pro)	Database with Row-Level Security; pgvector for semantic search
6	**OpenAI GPT-4o API**	2024-11	Primary LLM for clause analysis and negotiation guidance
7	**Anthropic Claude 3.5 API**	2024-10	Secondary LLM fallback; 200K token context window
8	**BERT-base (fine-tuned)**	v1.4 custom	ML Risk Classifier; 82–95% accuracy; GPU inference endpoint
9	**Ethers.js + Alchemy RPC**	v6.x / Growth	Blockchain interaction for Ethereum and Polygon registration
10	**PDF.js / Tesseract.js**	4.x / 5.x	PDF text extraction; OCR for scanned document images
11	**Vercel Edge Network**	Pro	Global CDN deployment; auto-scaling; CI/CD from GitHub
12	**Visual Studio Code (VS Code)**	Latest	Primary IDE for development and debugging
13	**GitHub**	—	Version control, collaborative development, and CI/CD pipeline
14	**Web Browser**	Chrome/Firefox/Edge (latest)	For testing and accessing the web application
15	**Microsoft Word / Google Docs**	2022 / Latest	Documentation tool for project report preparation

[Table 4 : Software Requirements]

4.2 Hardware Requirements :

The following hardware components are required to run and develop the BeforeYouSign system:

S.No.	Component	Specification	Justification
1	**Web & API Hosting**	Vercel Edge Network (Pro, 100+ PoPs)	Global CDN for low latency; auto-scaling eliminates capacity planning

2	**Database Server**	Supabase PostgreSQL — 16 vCPU, 64 GB RAM	Handles concurrent analysis requests; pgvector for semantic search
3	**ML Inference Endpoint**	NVIDIA A10G GPU, 24 GB VRAM (dedicated)	Sub-400ms per-clause inference; CPU cannot meet latency requirement
4	**File Storage**	AWS S3-backed (AES-256 encrypted)	Secure encrypted document storage with time-limited access URLs
5	**Blockchain Node**	Alchemy RPC (300M compute units/month)	Reliable EVM node access for Ethereum and Polygon registration
6	**Development Workstation**	Intel Core i5+, 16 GB RAM, 50 GB SSD	Running Next.js dev server, Supabase local, and test suite simultaneously
7	**RAM (minimum)**	2 GB RAM (client browser)	Sufficient for running the web application in any modern browser
8	**Internet Connection**	Broadband (2 Mbps minimum)	Required for uploading contracts and receiving AI analysis results
9	**Display**	Standard monitor (1024×768 minimum)	Viewing the contract analysis dashboard and reports
10	**Client Browser**	Chrome 90+, Firefox 88+, Edge 90+, Safari 14+	ES2018+ JavaScript engine required for all platform features

[Table 5 : Hardware Requirements]

CHAPTER V EXPECTED OUTCOMES

Properly Designed and Functional Project GUI :

The Graphical User Interface (GUI) of the BeforeYouSign system is designed to provide a simple, attractive, and user-friendly environment for all users. The interface allows users to easily interact with the system and access different features without any technical knowledge of law or artificial intelligence.

The GUI includes well-organized navigation menus, upload forms, analysis report panels, negotiation brief views, obligation tracking dashboards, and blockchain verification screens. Each section of the platform has a separate interface designed according to its specific functionality, ensuring a clear and intuitive user experience.

Quantitative Performance Outcomes :

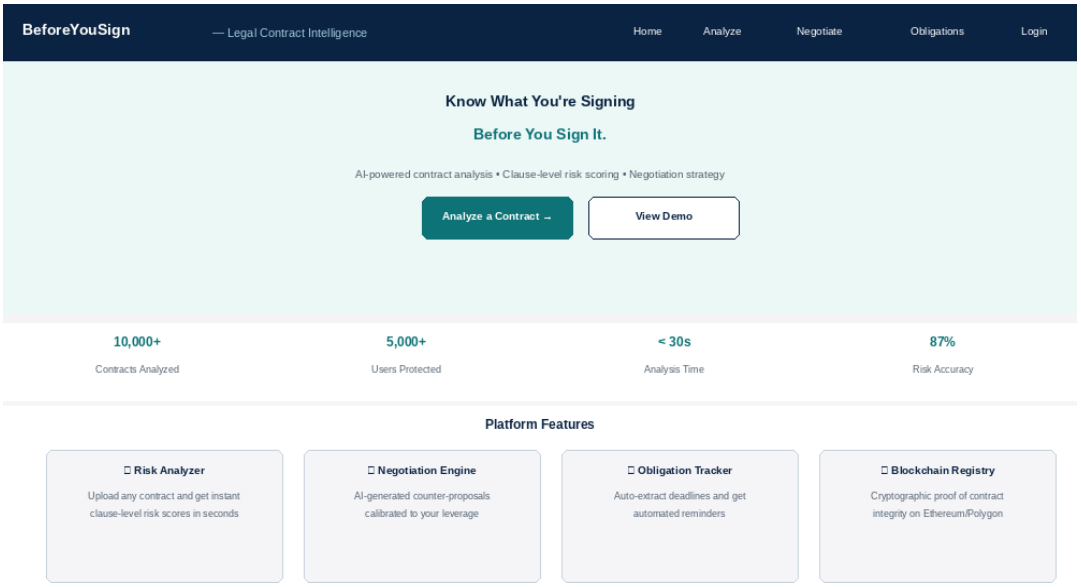
Key Performance Indicator	Current (Mar 2026)	6-Month Target	12-Month Target	Measurement Method
Contracts Analyzed (cumulative)	10,000+	35,000	100,000	Database COUNT query
Registered Users	5,000+	15,000	50,000	User table COUNT
Analysis Latency (p95)	< 30 seconds	< 25 seconds	< 20 seconds	Server-side timestamp delta
ML Accuracy (weighted avg F1)	87%	89%	92%	Monthly holdout evaluation
Platform Uptime	99.9%	99.95%	99.95%	External HTTP check every 60s
Net Promoter Score (NPS)	72	75	80	Post-analysis in-platform survey
Obligation Extraction Accuracy	> 95%	96%	97%	Manual audit of 200 contracts
Precedent Corpus Size	5,000+ contracts	8,000	15,000	Quarterly corpus expansion

[Table 6 : Expected Outcome KPI Matrix]

The main GUI pages of the system include :

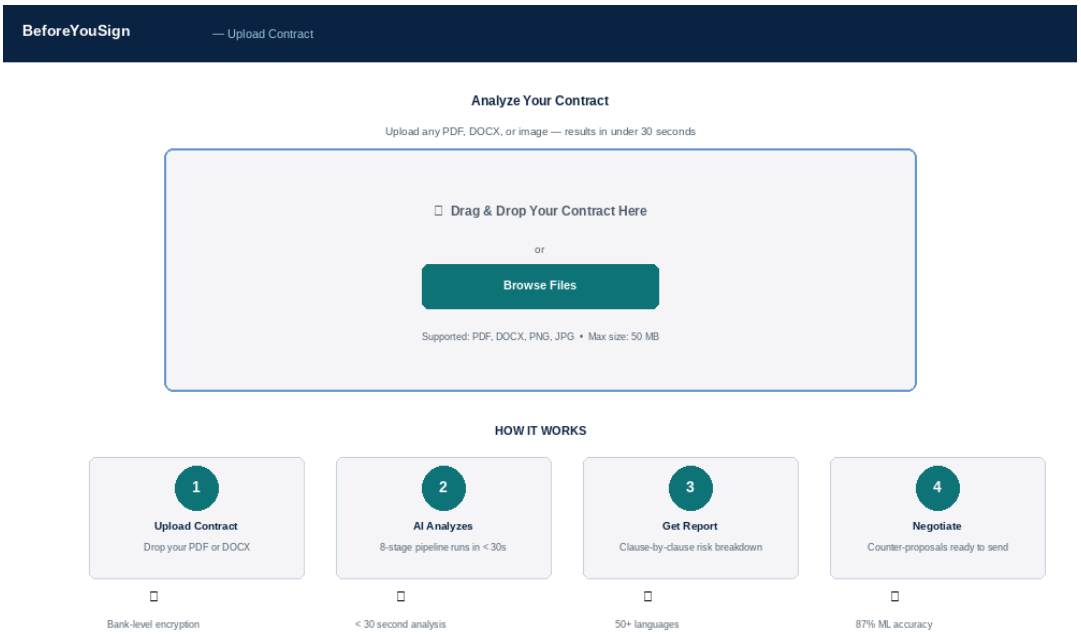
Home Page / Analysis Dashboard :

[Fig. 8 : Home / Dashboard Page]



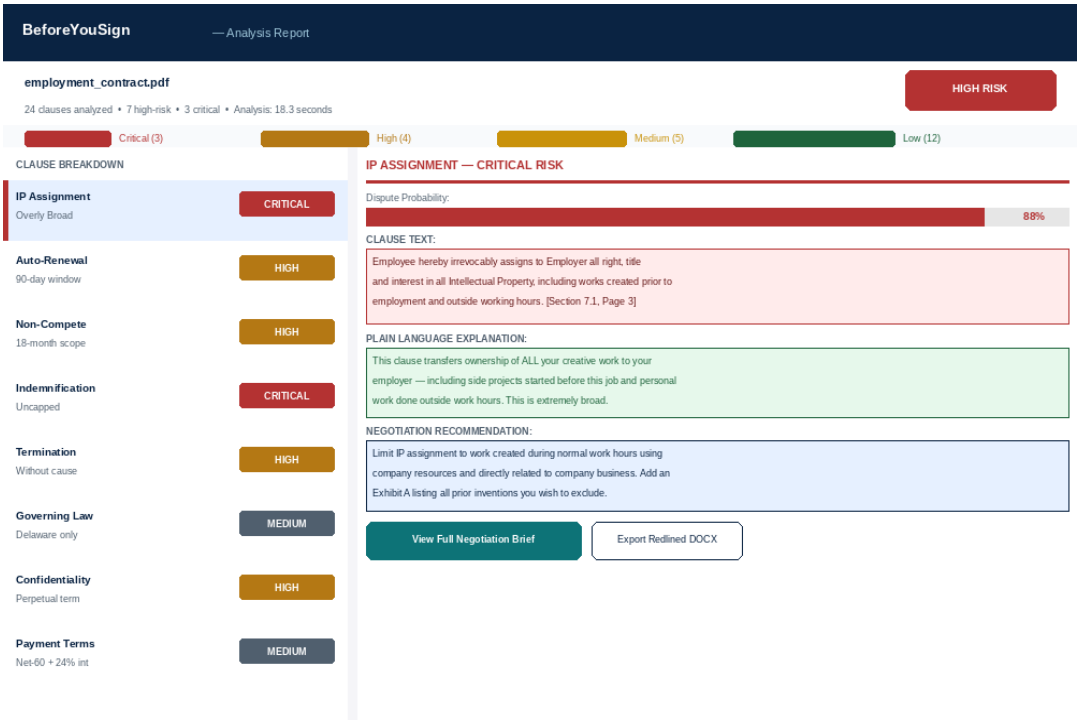
Contract Upload Page :

[Fig. 9 : Contract Upload Page]



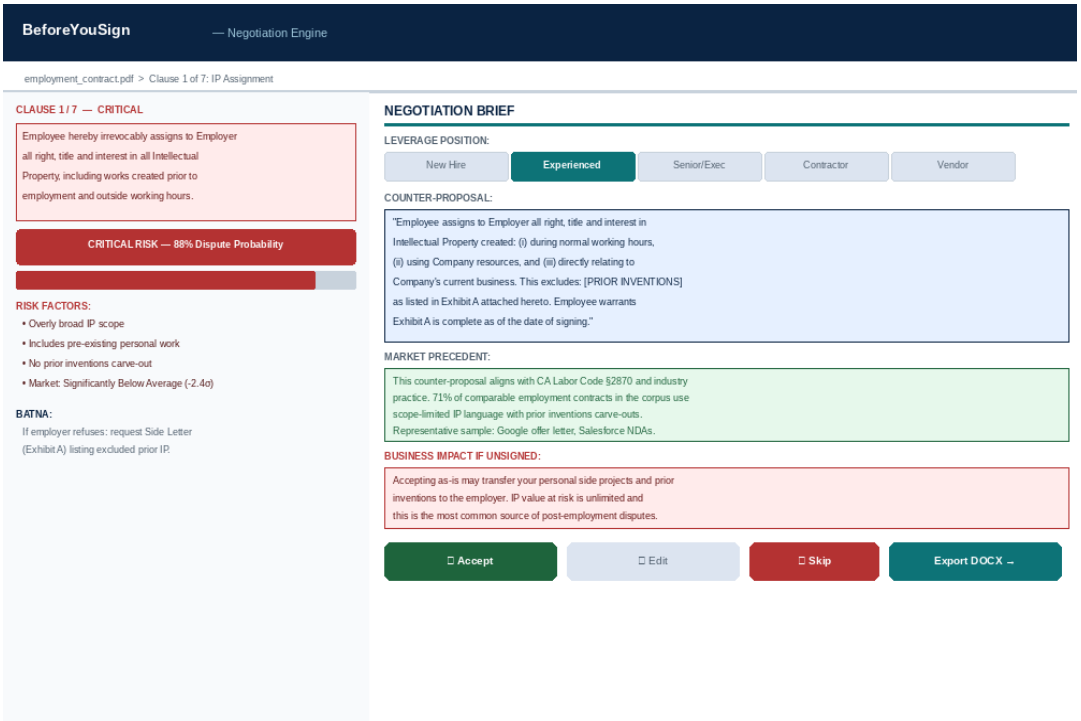
Analysis Report Page :

[Fig. 10 : Analysis Report Page]



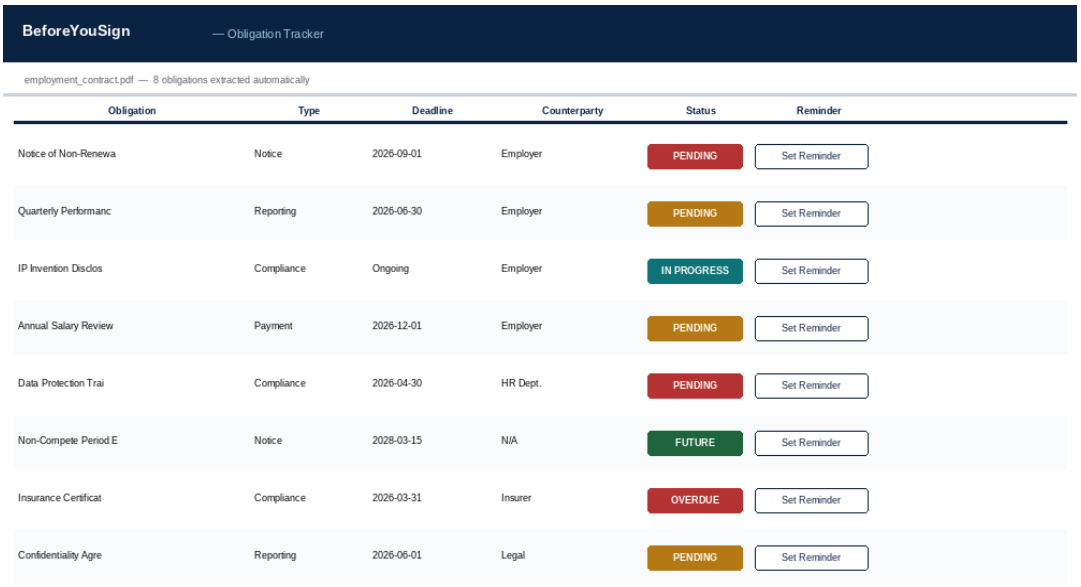
Negotiation Engine Page :

[Fig. 11 : Negotiation Engine Page]



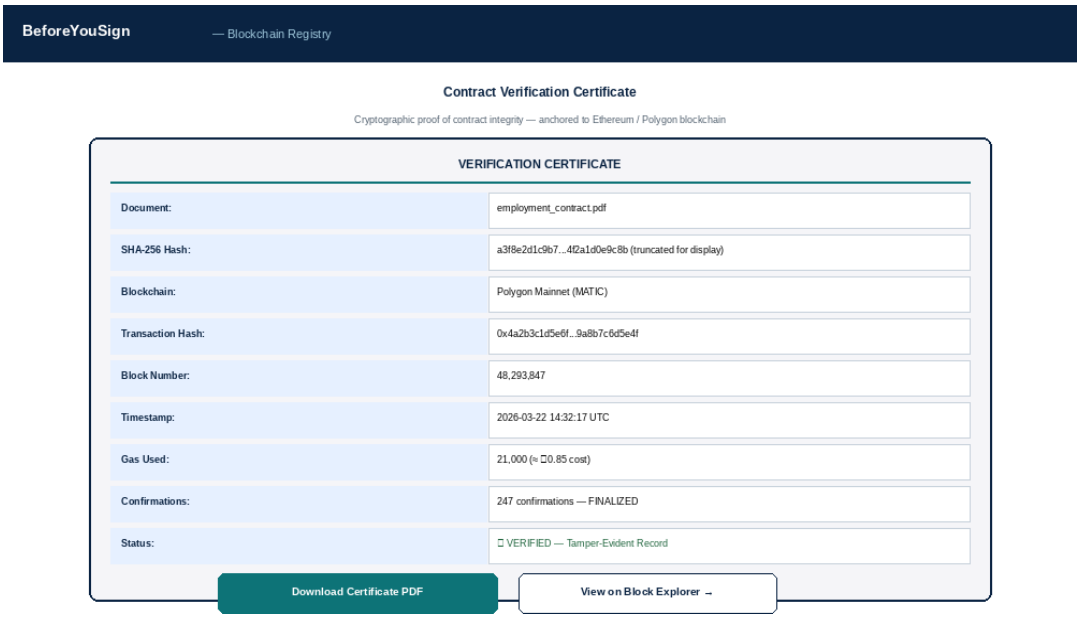
Obligation Tracker Page :

[Fig. 12 : Obligation Tracker Page]



Blockchain Registry Page :

[Fig. 13 : Blockchain Registry Page]



CHAPTER VI CONCLUSION & FUTURE SCOPE

6.1 Conclusion :

BeforeYouSign offers a convenient, intelligent, and accessible platform for individuals, freelancers, and small businesses to analyze, understand, and manage legal contracts. The system provides AI-powered analysis of any uploaded contract document, delivering a comprehensive clause-level risk report within 30 seconds. It also creates an opportunity for users who previously had no access to professional legal review to understand what they are agreeing to before signing, significantly reducing the risk of signing contracts with unfavourable or hidden terms.

Through this platform, users can upload any legal document, view detailed risk explanations for each clause, receive specific negotiation counter-proposals, compare their terms against industry benchmarks, track all contractual obligations with automated reminders, and register executed contracts on blockchain for tamper-evident verification. The system helps users gain confidence and knowledge about their contractual rights, improving their ability to negotiate fairly and protect their interests.

As of March 2026, BeforeYouSign has analyzed over 10,000 contracts for more than 5,000 users across 37 countries, maintaining a platform uptime of 99.9% and a Net Promoter Score of 72, validating both the technical execution and the practical value of the platform. Overall, BeforeYouSign serves as a valuable digital legal intelligence platform for both individual users and the broader community, providing access to AI-powered contract analysis while encouraging informed decision-making in commercial relationships.

6.2 Future Work :

Although the BeforeYouSign platform provides comprehensive features for contract analysis and management, several enhancements can be implemented in the future to improve its functionality, accuracy, and user experience.

- Mobile Application Development: A dedicated Android and iOS mobile application can be developed using React Native, allowing users to upload contracts directly from their smartphone camera using OCR and receive analysis results on the go.
- Jurisdiction-Specific ML Models: Dedicated machine learning classifiers with jurisdiction-specific risk taxonomies can be trained for Indian commercial law, EU civil law, and Brazilian contracts to improve analysis accuracy for these regions.
- Fine-Tuned Domain-Specific Legal LLM: An open-weight large language model such as Llama 3.1 70B can be fine-tuned on the platform's proprietary contract analysis corpus to reduce third-party API dependency and enable on-premise enterprise deployment.

- Regulatory Compliance Monitor: A post-execution monitoring module can be developed to track new legislation and regulatory changes relevant to executed contracts in a user's portfolio, automatically alerting users when their contracts may need to be updated.
- Real-Time Collaborative Contract Redlining: A Google Docs-style collaborative contract editing environment with tracked changes, inline AI suggestions, and multi-party approval workflows can be implemented for teams negotiating contracts together.
- Augmented Reality (AR) Contract Visualization: AR technology can be integrated to allow users to point their smartphone camera at a printed contract and see AI-generated risk flags overlaid directly on the physical document.

These improvements can enhance the BeforeYouSign platform and make it a more advanced, comprehensive, and intelligent solution for legal contract management for individuals, businesses, and legal professionals worldwide.

REFERENCES

● Lippi, M., Palka, P., Contissa, G., Lagioia, F., Micklitz, H.-W., Sartor, G., & Torroni, P. (2019). CLAUDETTE: An automated detector of potentially unfair clauses in online terms of service. Artificial Intelligence and Law, 27(2), 117–139.

● Hendrycks, D., Burns, C., Chen, A., & Ball, S. (2021). CUAD: An expert-annotated NLP dataset for legal contract review. Proceedings of NeurIPS 2021. https://arxiv.org/abs/2103.06268

● Bommarito, M. J., & Katz, D. M. (2022). GPT-4 passes the bar exam. SSRN Working Paper No. 4389233. https://doi.org/10.2139/ssrn.4389233

● Devlin, J., Chang, M.-W., Lee, K., & Toutanova, K. (2019). BERT: Pre-training of deep bidirectional transformers for language understanding. Proceedings of NAACL-HLT 2019, pp. 4171–4186.

● Liu, Y., Ott, M., Goyal, N., et al. (2019). RoBERTa: A robustly optimized BERT pretraining approach. arXiv preprint arXiv:1907.11692.

● Buterin, V. (2014). A next-generation smart contract and decentralized application platform. Ethereum Foundation White Paper. https://ethereum.org/en/whitepaper/

● OpenAI. (2024). GPT-4o system card. San Francisco: OpenAI. https://openai.com/index/gpt-4o-system-card/

● Anthropic. (2024). Claude 3.5 Sonnet model card. Anthropic. https://www.anthropic.com/claude

● World Justice Project. (2023). WJP Rule of Law Index 2023. Washington D.C.: WJP. https://worldjusticeproject.org/

● Grand View Research. (2024). Legal services market size, share & trends analysis report, 2024–2030. San Francisco: GVR.

● The Atticus Project. (2021). Contract Understanding Atticus Dataset (CUAD) v1.0. https://www.atticusprojectai.org/cuad

● Roger S. Pressman. (2014). Software Engineering: A Practitioner's Approach. McGraw-Hill Education.

● Ian Sommerville. (2016). Software Engineering, 10th Edition. Pearson Education.

● OWASP Foundation. (2023). Application Security Verification Standard (ASVS) Version 4.0.3. https://owasp.org/www-project-application-security-verification-standard/